ENEE 132 — Basic Electronics I (Semiconductor Devices)

Lecture 1 - Atomic Structure, Materials & Covalent Bonding

2 June 2026

About the Instructor

- **Dr. Roger Ahiadormey**
  - Research Scientist, CSIR-INSTI
  - PhD in Electronics Engineering
- **Research Interests**
  - 5G/6G
  - Internet-of-Things (IoT)
  - Artificial Intelligence (AI)
  - Tiny Machine Learning (TinyML)
- **Contact**
  - Email: rogerkwao@gmail.com
  - Website: sites.google.com/view/rogerkwao/home

Course Overview

- **Course Objectives**
  - Master principles of semiconductor devices that comprise today's integrated circuits
  - Cover basic electronic devices and circuits, including troubleshooting and system applications

- **Course Approach**
  - Theory grounded in physical intuition, not just formulas
  - Connections from atoms all the way up to working circuits
- **Course Outcome**
  - You will read a circuit schematic and predict its behaviour
  - You will choose the right device for the right application

Required & Recommended Reading

- **Primary Source**
  - Course Lecture Notes — the main reference for all lectures and assessments
- **Additional Reading**
  - Schultz, M. E. (2015). Grob's Basic Electronics (12th ed.). McGraw-Hill.
  - Buchla, T. L. (2009). Electronics Fundamentals: Circuits, Devices & Applications (8th ed.). Pearson.
  - Neamen, D. A. (2011). Semiconductor Physics and Devices: Basic Principles (4th ed.). McGraw-Hill.
  - Lee, S. M.-K. (2012). Semiconductor Devices: Physics and Technology (3rd ed.). Wiley.
  - Floyd, T. L., Buchla, D. M., & Wetterling, S. (2018). Electronic Devices (Conventional Current Version) (10th ed.). Pearson.
  - Fiore, J. M. Semiconductor Devices: Theory and Application (2nd ed.). Open-access (dissidents).
- **Useful Online Resources**
  - Multisim – for simulation device operation
- **Reading Expectations**
  - Skim the relevant section of the Lecture Notes before lecture; reread after
  - Do all end-of-section CHECK-UP questions

Assessment & Grading

- **Continuous Assessment — 40 percent**
  - Homework problems
  - Two quizzes
- **Final Examination — 60 percent**
  - Mix of conceptual questions and numerical analysis
- **Class Website**
  - https://github.com/rogerkwao1/Basic-Electronics_Semiconductor-Devices

Why Study Semiconductor Devices?

- **They Are Everywhere**
  - Every phone, laptop, car, power supply and IoT sensor relies on semiconductors
  - A modern CPU contains tens of billions of transistors
- **Career Relevance**
  - Electronics, telecommunications, embedded systems, power systems
  - Bridge between software engineering and physical hardware

Course Roadmap — 9 Weeks

Week	Title	Pages
Foundations (Weeks 1–2)		
Week 1	***Atomic Structure, Materials & Covalent Bonding — today***	pp. 1–15
Week 2	Conduction, Doping & the PN Junction	pp. 16–27
The Diode (Weeks 3–7)		
Week 3	The Diode — Operation, Characteristics & Models	pp. 28–46
Week 4	Rectifiers (Half-Wave, Full-Wave, Bridge)	pp. 47–65
Week 5	Power-Supply Filters & Voltage Regulators	pp. 66–74
Week 6	The Zener Diode & Applications	pp. 75–89
Week 7	Special-Purpose Diodes	pp. 90–116
The Transistor (Weeks 8–9)		
Week 8	BJT Operation & Characteristics	pp. 117–135
Week 9	BJT as Amplifier & as Switch	pp. 136–144

Today's Agenda — Lecture Notes pp. 1–15

- **Part 1 — Atomic Structure**
  - Bohr model, subatomic particles, the periodic table
- **Part 2 — Shells, Orbits & Energy**
  - Electron shells, the $2n^2$ rule, free electrons
- **Part 3 — Valence Electrons & Ionization**
  - Valence count, ions, the electron-volt
- **Part 4 — Quantum Model**
  - Why the Bohr model is incomplete
- **Part 5 — Materials Used in Electronics**
  - Conductors, insulators, semiconductors
- **Part 6 — Atomic Structure of Semiconductors**
  - Silicon & germanium, covalent bonding, the silicon crystal

Today's Learning Objectives

- **By the End of This Lecture, You Will Be Able To**
    - Describe the structure of an atom (nucleus, shells, valence electrons)
    - Apply the $2n^2$ rule to predict shell capacity
    - Explain ionization and distinguish positive from negative ions
    - Classify any material as conductor, insulator, or semiconductor from its valence count
    - Identify silicon and germanium as Group IV semiconductors
    - Explain covalent bonding and describe the silicon crystal lattice

Part 1 — Atomic Structure

What Is an Atom?

- **Definition**
  - Smallest particle of an element that retains its properties
- **Composition**
  - Protons and neutrons in the nucleus
  - Electrons orbiting in shells
- **Uniqueness**
  - Each of the 118 known elements has a unique atomic structure
  - All atoms of an element share the same proton count

The Bohr Model

- **Niels Bohr's Proposal (1913)**
  - Electrons orbit the nucleus at discrete, allowed distances
  - Similar to planets orbiting the sun — hence "planetary model"
- **Why We Use It**
  - Easy to visualise and reason about
  - Sufficient for almost all of our electronics analysis
- **Its Limits**
  - A modern quantum-mechanical model is more accurate
  - But the quantum model is much harder to picture
- **Rule of Thumb**
  - Use Bohr for intuition; remember the quantum model is more correct

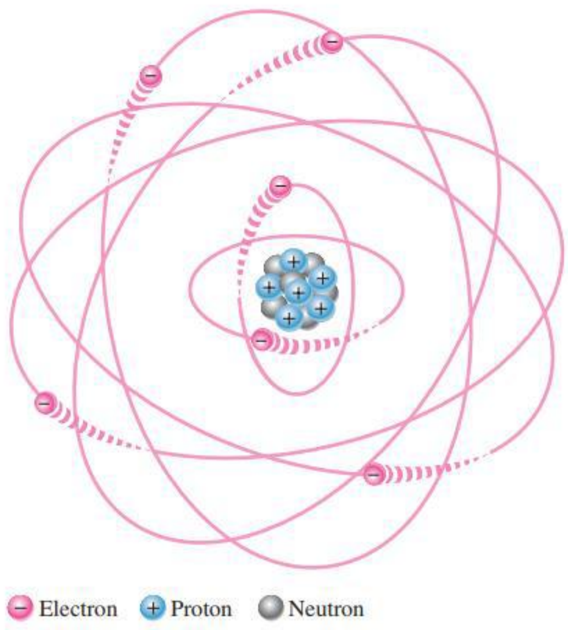


Fig 1.1 — Bohr model of an atom

The Three Subatomic Particles

- **Proton (+1)**
  - Lives in the nucleus
  - Number of protons = atomic number
- **Neutron (0)**
  - Lives in the nucleus
  - Neutral; adjusts atomic mass
- **Electron (−1)**
  - Orbits in shells around the nucleus
  - Mass roughly 1/1836 of a proton

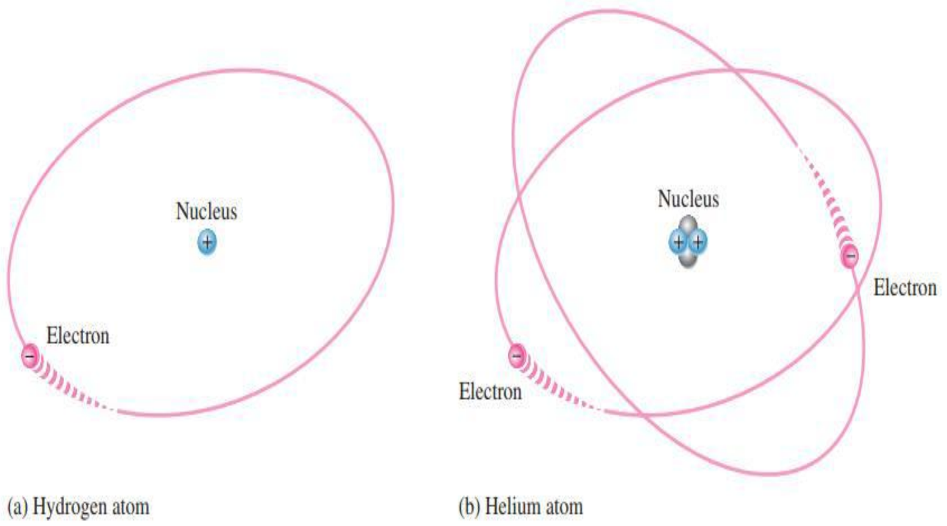


Fig 1.2 — Two simple atoms (hydrogen and helium)

Atomic Number

- **Definition**
  - Atomic number = number of protons in the nucleus
  - In a neutral atom, the atomic number also equals the number of electrons
- **Why It Matters**
  - The atomic number uniquely identifies the element
  - For Hydrogen, atomic number = 1, For Helium, atomic number = 2, For Carbon, atomic number = 6, For Silicon, atomic number = 14
- **On the Periodic Table**
  - Elements are ordered by the atomic number
  - Atomic mass and electron configuration follow predictably

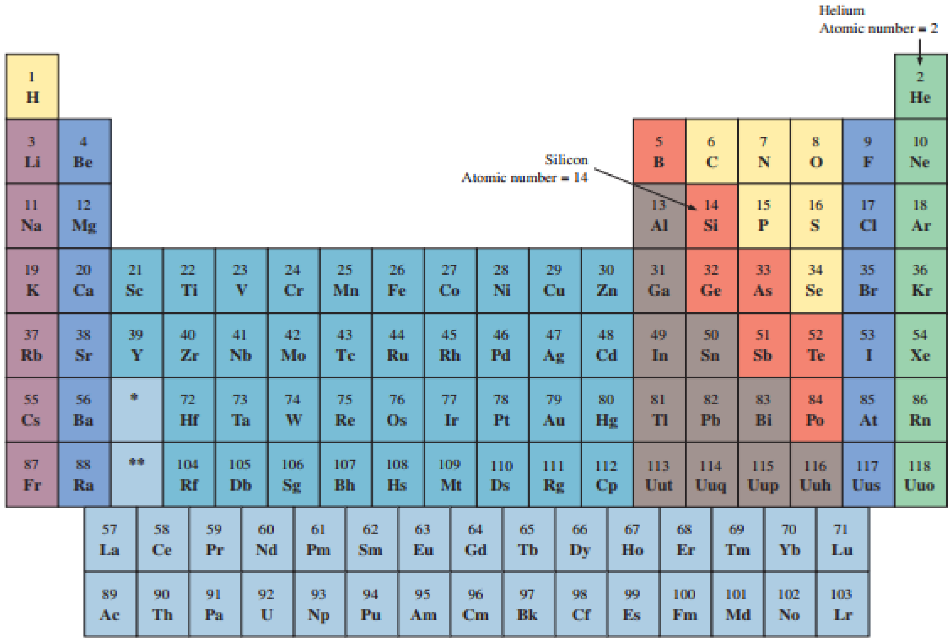


Fig 1.3 — The periodic table of the elements

Part 2 — Shells, Orbits & Energy

Electron Shells

- **Shells = Energy Groups**
  - Allowed orbits cluster into shells labelled 1, 2, 3, 4, …
  - Shell 1 is closest to the nucleus, shell 7 is farthest
  - Energy increases with distance — outer electrons are higher-energy and more easily freed
- **Each Shell Has a Maximum Capacity**
  - Once a shell fills, additional electrons go to the next shell out
- **Why Shells Exist**
  - Only discrete electron energies exist, so orbits cluster into separate shells
  - Each shell holds only a fixed maximum number of electrons (the $2n^2$ rule)

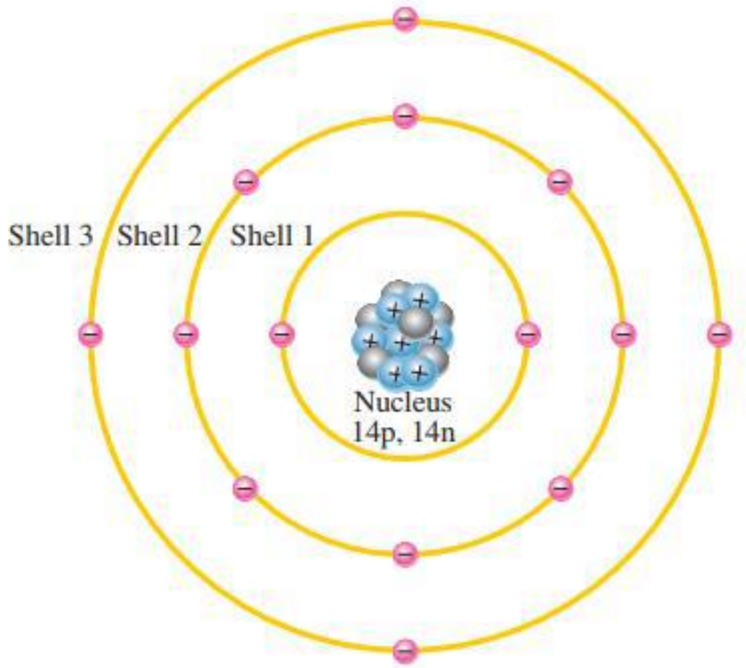


Fig 1.4 — Bohr model of the silicon atom

Maximum Electrons per Shell — The $2n^2$ Rule

- **The Formula**
  - Max electrons in shell n = $2 \times n^2$
- **Applied**
  - Shell 1: $2 \times 1^2 = 2$ electrons
  - Shell 2: $2 \times 2^2 = 8$ electrons
  - Shell 3: $2 \times 3^2 = 18$ electrons
  - Shell 4: $2 \times 4^2 = 32$ electrons

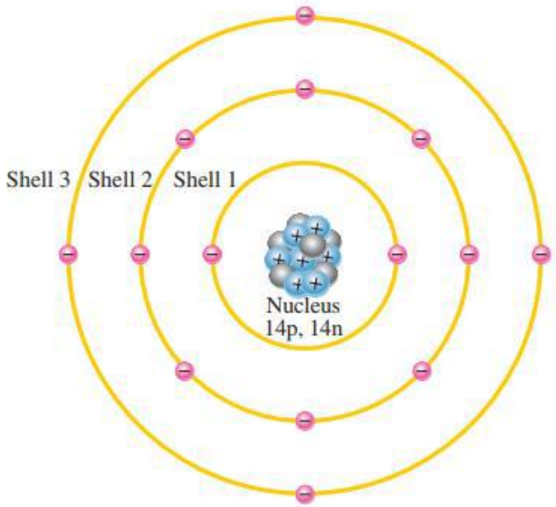


Fig 1.4 — Bohr model of the silicon atom

Part 3 — Valence Electrons & Ionization

Valence Electrons — Definition

- **Definition**
  - Valence electrons live in the outermost occupied shell — the valence shell
- **Their Properties**
  - Highest energy of any electrons in the atom
  - Weakest binding to the nucleus
  - Only electrons that participate in chemical bonding
- **Examples**
  - Hydrogen: 1 valence electron
  - Carbon, silicon, germanium: 4 valence electrons each

Visualizing Valence — Silicon Revisited

- **Silicon's Valence Shell**
  - Shell 3 holds 4 valence electrons (highlighted orange)
  - The 10 inner electrons (shells 1 & 2) are tightly bound — irrelevant for electronics
- **What is Important**
  - Only the 4 outer electrons participate in bonding and current
  - Doping works by tampering with this outer shell
- **Mental Model**
  - Think of silicon as a +4 core surrounded by 4 loose electrons
  - Inner shells are invisible for circuit purposes

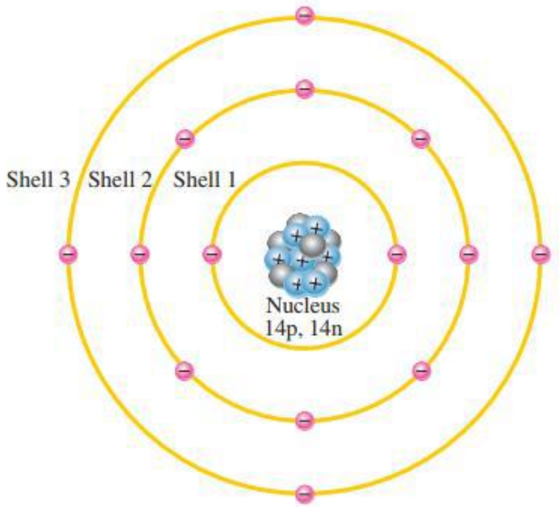


Fig 1.4 — Bohr model of the silicon atom

Ionization

- **Definition**
  - The process by which an atom gains or loses one or more electrons
  - Result: atom acquires a net non-zero charge
- **Energy Requirement**
  - Removing an electron requires energy — the ionization energy
  - Inner-shell electrons need far more energy than valence
  - Low ionization energy → easily forms free electrons → tends to be a conductor
- **Result — An Ion**
  - An atom with net charge
  - Positive ion if it lost electrons
  - Negative ion if it gained electrons

Positive Ions

- **Positive Ion Formation**
  - An atom loses one or more electrons
  - Now has more protons than electrons → net positive charge
- **Example — Hydrogen (Atomic number = 1)**
  - A neutral hydrogen atom (H) has one valence electron
  - It loses that electron to become a positive ion, H^+
  - The escaped valence electron is called a free electron

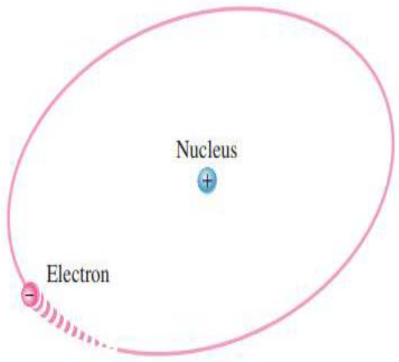


Fig 1.5 — Ionization of hydrogen (positive ion)

Negative Ions

- **Negative Ion Formation**
  - An atom gains one or more electrons
  - Now has more electrons than protons → net negative charge
- **Example — Chlorine (Atomic number = 17)**
  - A free electron can be captured by a chlorine atom
  - It becomes a negative ion, Cl^{-}
  - More stable because it now has a filled outer shell

Part 4 — Quantum Model

Why Bohr Model Isn't Enough

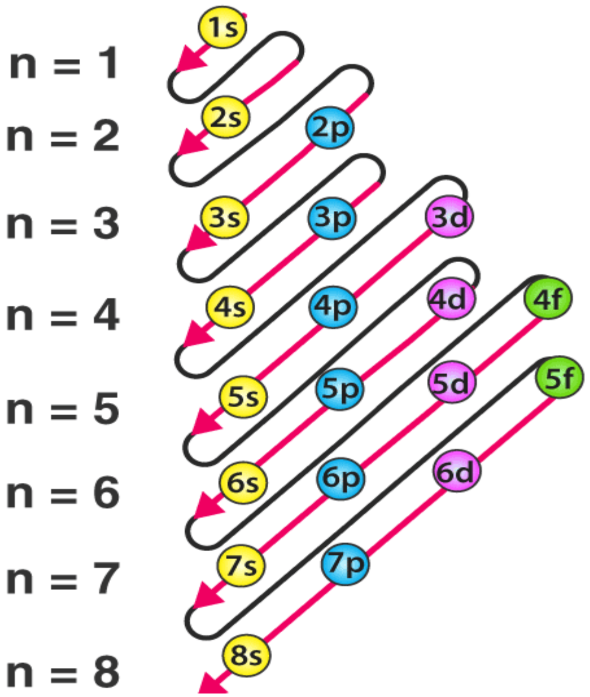
- **Bohr's Limitations**
  - Cannot explain fine details of atomic spectra
  - Treats electrons as point particles in fixed orbits — wrong
  - Heisenberg uncertainty rules out exact orbits anyway
- **What the Quantum Model Says**
  - Electrons exist in probability clouds called orbitals
  - Each orbital has a specific shape (s, p, d, f) and energy
  - Energy is still quantized (discrete levels)
- **When to Use Which**
  - Bohr — device-level intuition and circuit reasoning
  - Quantum — precise spectra, quantum mechanics, advanced physics

Three Quantum Principles

- **Heisenberg Uncertainty Principle**
  - You cannot know both an electron's exact position and exact momentum simultaneously
  - The more precisely you know one, the less you know the other
- **Wave-Particle Duality**
  - Electrons behave like waves AND particles, depending on experiment
  - Wave nature gives rise to the orbital shapes (s, p, d, f)
- **Superposition Principle**
  - Until observed, a particle exists in all of its possible states simultaneously
  - Illustrated by Schrödinger’s cat — both alive and dead until an observation is made

Orbitals & Electron Configuration

- **Subshells Called Orbitals**
  - Each shell holds up to four subshells called orbitals: s, p, d, and f
  - Capacities: s holds 2, p holds 6, d holds 10, and f holds 14 electrons
- **The Electron Configuration Table**
  - Lists each shell, its orbitals, and the number of electrons in each
  - Notation: shell number, orbital letter, then an exponent for the electrons
  - The next slides give the tables for nitrogen and for silicon



Subshell (orbital) filling order, n = 1 to 8

The Electron Configuration Table — Nitrogen

Each atom can be described by an electron configuration table. For nitrogen (Atomic number = 7):

NOTATION	EXPLANATION
$\boldsymbol{1s^2}$	2 electrons in shell 1, orbital *s*
$\boldsymbol{2s^2\ \ 2p^3}$	5 electrons in shell 2: 2 in orbital *s*, 3 in orbital *p*

How to read it: the first full-size number is the shell (energy level), the letter is the orbital, and the exponent is the number of electrons in the orbital.

EXAMPLE 1–1 — Silicon (Atomic number = 14)

- **Problem**
  - Using the atomic number from the periodic table, describe a silicon (Si) atom using an electron configuration table.
- **Solution**
  - The atomic number of silicon is 14, so there are 14 protons — and 14 electrons in the neutral atom
  - Shells hold up to 2 (shell 1), 8 (shell 2), 18 (shell 3); so silicon has 2, 8, and 4 — total 14
  - Silicon's electron configuration table (Table 1.2) is shown below

NOTATION	EXPLANATION
$1s^2$	2 electrons in shell 1, orbital *s*
$2s^2\ 2p^6$	8 electrons in shell 2: 2 in orbital *s*, 6 in orbital *p*
$3s^2\ 3p^2$	4 electrons in shell 3: 2 in orbital *s*, 2 in orbital *p*

Orbital Shapes — s and p Orbitals

- **s-orbitals**
  - Shaped like spheres with the nucleus at the centre
  - Shell 1 is a single sphere; for shells 2 and up, each s-orbital is nested spherical shells
- **p-orbitals**
  - Two ellipsoidal lobes meeting at the nucleus — a dumbbell shape
  - The three p-orbitals in a shell are at right angles, on the x, y and z axes
- **Example — Sodium (Na, 11 electrons)**
  - Figure shows the orbitals in 3-D, with the number of electrons in each

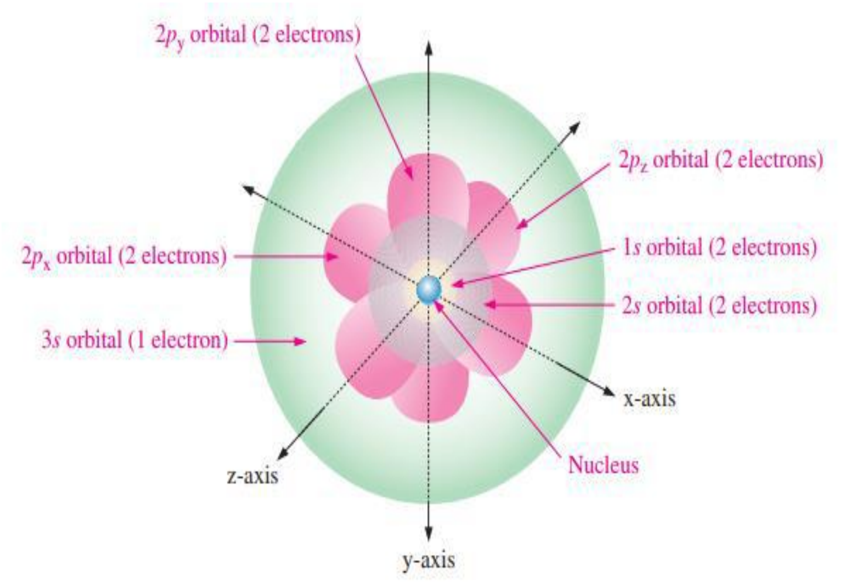


Fig 1.6 — 3-D quantum model of the sodium atom

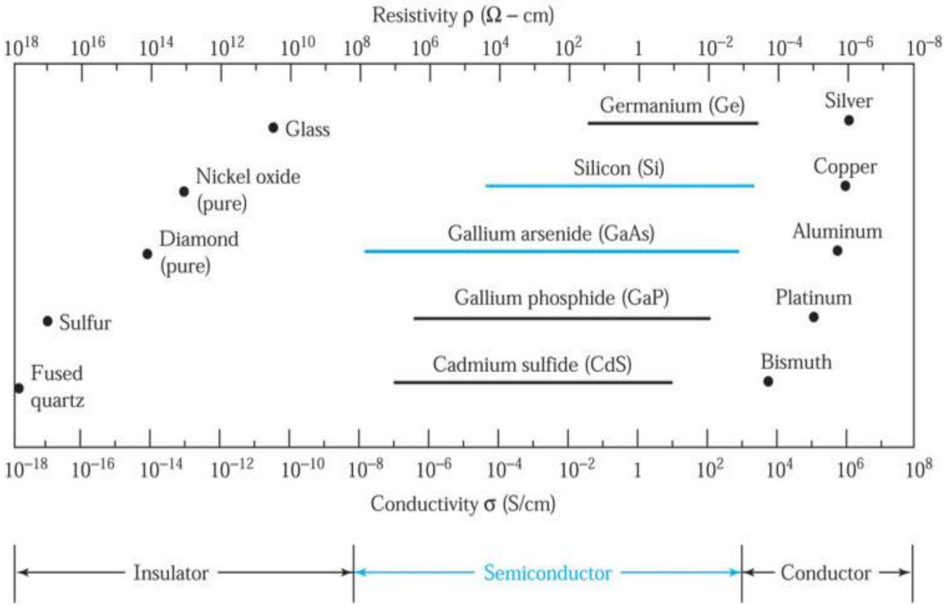
Part 5 — Materials Used in Electronics

Three Classes of Materials

- **Conductor**
  - Allows current to flow easily at room temperature
  - Has many free electrons (low resistivity)
  - Examples: copper, aluminium, silver, gold
- **Insulator**
  - Blocks current almost completely
  - Has very few free electrons (high resistivity)
  - Examples: rubber, glass, plastic, mica
- **Semiconductor**
  - Sits between conductor and insulator
  - Conducts a little and the amount can be engineered
  - Examples: silicon, germanium, gallium arsenide

Conductors

- **Definition**
  - A material in which current flows freely at room temperature
  - Typical resistivity: ~10^{-8} Ω·m
- **Why They Conduct**
  - Valence electrons are weakly bound, so they easily become free electrons
  - A small applied voltage drives a large current
- **Common Examples**
  - Copper (Cu) — everyday wiring
  - Aluminium (Al) — overhead transmission lines
  - Silver (Ag), gold (Au) — premium contacts



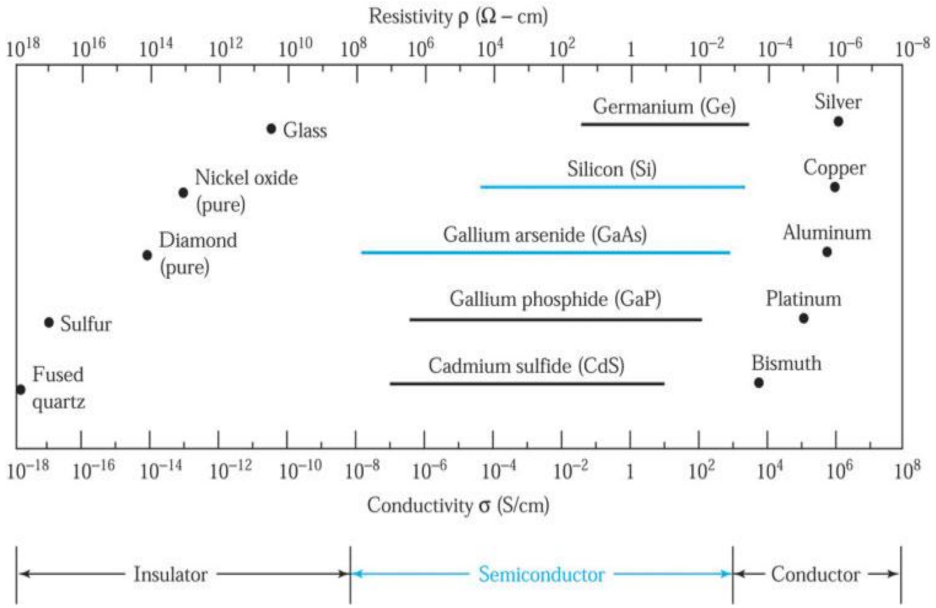
Resistivity and conductivity of common materials

Atomic Structure of a Conductor

- **Key Feature**
  - The best conductors have just one valence electron (e.g. copper)
  - Those valence electrons sit far from the nucleus and feel weak attraction
- **Copper (Atomic number=29)**
  - Configuration: 2-8-18-1 → just one valence electron
  - That single outer electron is essentially free at room temperature
- **Result**
  - A copper crystal hosts an "electron sea" — a cloud of mobile electrons shared across the lattice
  - Apply a voltage and the cloud drifts, producing current

Insulators

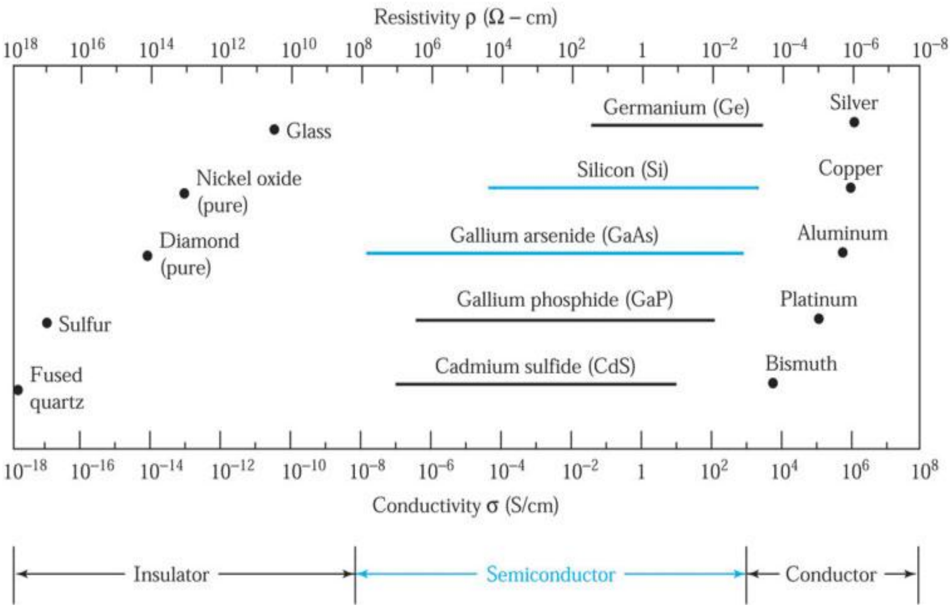
- **Definition**
  - A material that blocks current — almost no conduction at room temperature
  - Typical resistivity: 10^{10} Ω·m or higher
- **Why They Block**
  - Valence electrons are tightly bound in shared bonds
  - No free electrons available to carry current
- **Common Examples**
  - Rubber — wire insulation
  - Glass and porcelain — high-voltage insulators
  - Plastic, mica, dry wood



Resistivity and conductivity of common materials

Semiconductors

- **Definition**
  - A material that sits between conductor and insulator
  - Resistivity is in the middle range and — crucially — can be engineered
- **Why They Are Special**
  - Pure semiconductor conducts only weakly at room temperature
  - Adding tiny amounts of impurities ("doping") dramatically changes conductivity
- **Common Examples**
  - Single-element: antimony (Sb), arsenic (As), astatine (At), boron (B), polonium (Po), tellurium (Te), silicon (Si), germanium (Ge)
  - Compound: gallium arsenide (GaAs), indium phosphide (InP), gallium nitride (GaN), silicon carbide (SiC), silicon-germanium (SiGe)
  - Silicon is the most commonly used semiconductor



Resistivity and conductivity of common materials

Comparison at a Glance

- **Conductor vs Insulator vs Semiconductor**
  - Conductor: 1–3 valence electrons, low resistivity, many free electrons
  - Insulator: 5–8 valence electrons, very high resistivity, no free electrons
  - Semiconductor: exactly 4 valence electrons, intermediate resistivity, few free electrons (engineerable)
- **The Valence-Electron Rule**
  - The number of valence electrons predicts the material class
  - Four valence electrons → semiconductor (silicon, germanium, carbon in diamond form)
- **Why This Course Cares About Semiconductors**
  - Semiconductors are the only class whose behaviour we can tune by design
  - That tunability is what makes every electronic device possible

Band Gap — The Energy-Band View

- **From Levels to Bands**
  - In one atom, electrons sit at discrete energy levels
  - In a solid, billions of atoms merge these into energy bands
- **The Two Key Bands**
  - Valence band — highest filled band; electrons bound in covalent bonds
  - Conduction band — higher band; electrons here move freely and carry current
- **The Band Gap**
  - The forbidden energy region between the two bands
  - The difference in energy between the valence band and the conduction band
- **Gap Size Sets the Class**
  - Conductor: bands overlap — no gap
  - Insulator: very wide gap (diamond ≈ 5.5 eV)
  - Semiconductor: small gap (Si ≈ 1.1 eV) — modest heat frees some carriers

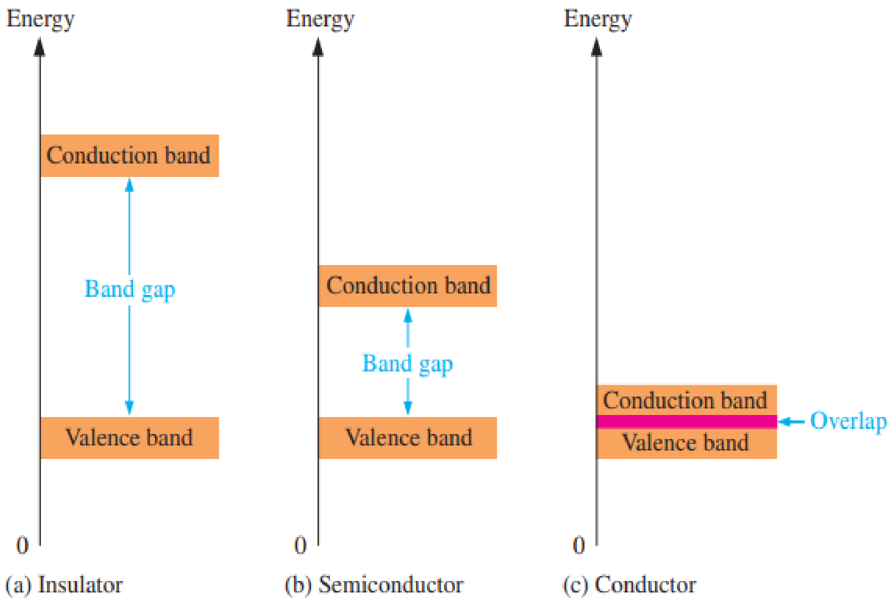


Fig 1.7 — Energy diagrams for the three types of materials

Comparison of a Semiconductor Atom to a Conductor Atom

- **The Two Atoms**
  - Silicon is a semiconductor; copper is a conductor
  - Their Bohr diagrams are shown in Fig 1.8 (right)
- **Core Charges**
  - Silicon core: +4 net charge (14 protons – 10 inner electrons)
  - Copper core: +1 net charge (29 protons – 28 inner electrons)
- **Remember**
  - The “core” is the entire atom except its valence electrons
  - The notes first show this with carbon: 6 protons – 2 inner electrons = a +4 core (4 valence electrons)

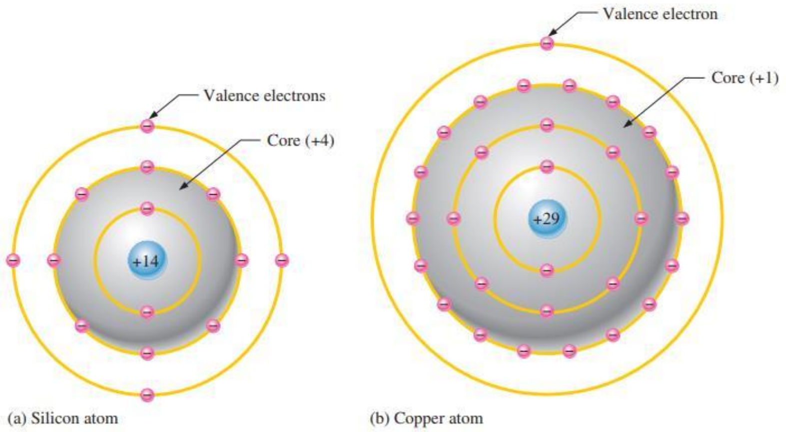


Fig 1.8 — Bohr diagrams of the silicon and copper atoms

Part 6 — Atomic Structure of Semiconductors

Silicon vs Germanium

- **What They Share**
  - Both Group IV: 4 valence electrons each
  - Both form the same diamond crystal lattice
  - Both behave identically in concept — they differ only in numbers
- **Where They Differ**
  - Bandgap: Si ≈ 1.1 eV; Ge ≈ 0.7 eV
  - Barrier potential (for diodes): Si ≈ 0.7 V; Ge ≈ 0.3 V
  - Native oxide: SiO_2 is excellent; GeO_2 is unstable
  - Stability: germanium's valence electrons escape more easily, so it is unstable at high temperature with excessive reverse current which is why silicon is preferred
- **In This Course**
  - We will use silicon as the default for every device unless stated otherwise
  - Germanium will appear when its lower barrier potential matters

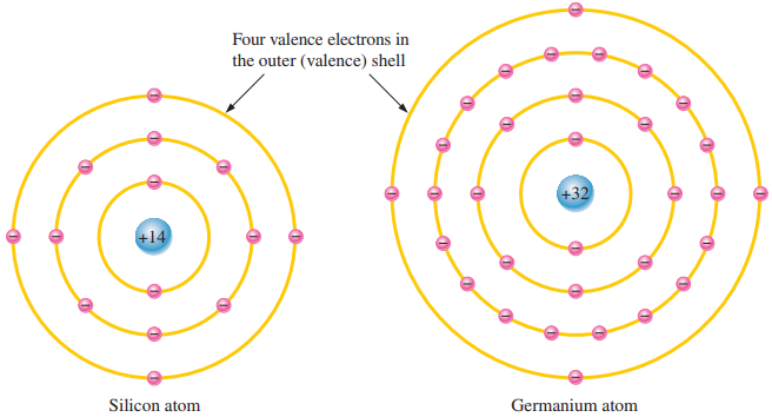


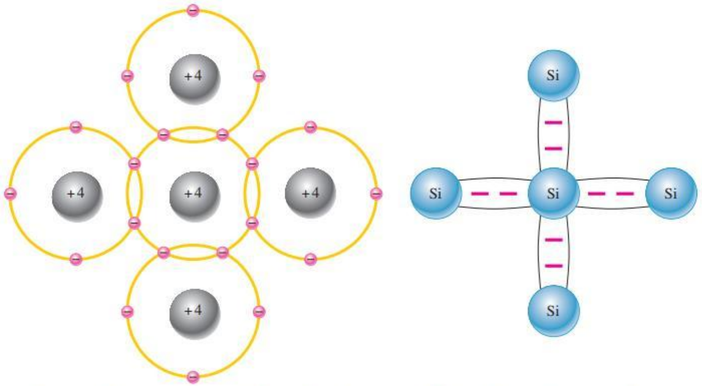
Fig 1.9 — Diagrams of the silicon and germanium atoms

Why Silicon Dominates

- **Materials Reasons**
  - Abundant and cheap (refined from sand)
  - Forms SiO_2 — a near-ideal insulator that sticks to silicon during processing
- **Manufacturing Reasons**
  - Photolithography on SiO_2 enables the entire integrated-circuit industry
  - Can be grown into large, defect-free single crystals (boules)
- **Practical Reasons**
  - Wider bandgap means devices stay reliable at higher temperatures
  - Si CMOS technology is the global standard — every laptop, phone, server

Covalent Bonding — One-Atom View

- **The Idea**
  - A covalent bond is a pair of electrons shared between two atoms
  - Each atom contributes one electron to the bond
- **Why Silicon Loves Covalent Bonds**
  - With 4 valence electrons, Si can form 4 covalent bonds
  - Sharing 4 outward and receiving 4 inward gives it a full 8-electron outer shell
- **Engineering Picture**
  - Each bond holds 2 electrons firmly between two atoms
  - No electrons are "free" — that is why pure silicon barely conducts



(a) The center silicon atom shares an electron with each of the four surrounding silicon atoms, creating a covalent bond with each. The surrounding atoms are in turn bonded to other atoms, and so on.

(b) Bonding diagram. The red negative signs represent the shared valence electrons.

Fig 1.10 — Covalent bonding in silicon

Covalent Bonding — Lattice View

- **Two-Dimensional Picture**
  - Each Si atom is shown with its 4 valence electrons
  - Lines between atoms represent shared electron pairs
- **Properties of the Lattice**
  - Repeating, regular pattern — every atom has identical neighbours
  - Strong, directional bonds — gives silicon its mechanical hardness
- **Reading the Diagram**
  - A neutral, undoped Si crystal has zero free carriers in this picture

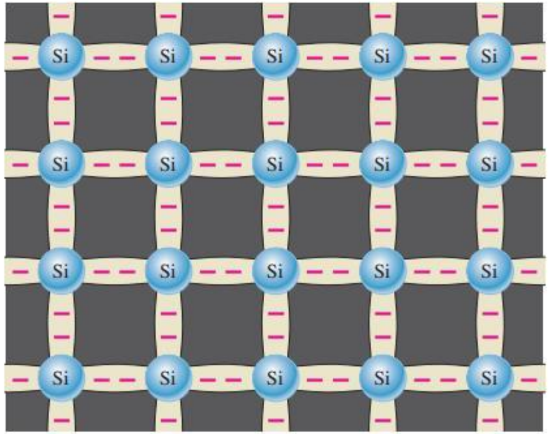


Fig 1.11 — A silicon crystal lattice

Worked Example — Counting Bonds in Silicon

- **Problem**
  - In a perfect silicon crystal, how many covalent bonds surround each atom?
- **Solution**
  - Each Si atom has 4 valence electrons → forms 4 bonds
  - Each bond is shared with one neighbouring atom
  - So each Si atom has 4 neighbours, each connected by 1 covalent bond
- **Bonus — Total Electrons in the Outer Shell**
  - Own contribution: 4 valence electrons
  - Plus: 4 electrons shared from neighbours
  - Apparent total around the atom: 8 — a full outer shell (the noble-gas configuration)

Free Electrons in Pure Silicon

- **At Absolute Zero (Theoretical)**
  - Every valence electron is locked in a covalent bond
  - Zero free electrons → silicon is a perfect insulator
- **At Room Temperature**
  - Thermal energy randomly breaks a small fraction of bonds
  - Each broken bond releases one free electron — and leaves a "hole" behind
- **Engineering Number**
  - Intrinsic silicon at 27 °C: about 1 free electron per 10^{12} atoms
  - That is why pure (intrinsic) silicon conducts only feebly

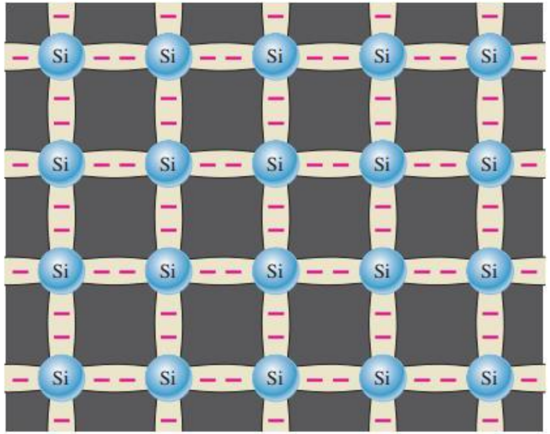


Fig 1.11 — A silicon crystal lattice

Key Takeaways from Today

- An atom is mostly empty space: a tiny nucleus with electrons in defined shells
- The $2n^2$ rule sets shell capacity; valence electrons determine behaviour
- Ionization removes or adds a valence electron (forms a positive or negative ion); energy is measured in eV
- Materials are conductors (1–3 valence e^-), insulators (5–8), or semiconductors (exactly 4)
- Silicon and germanium are the Group IV semiconductors; silicon dominates industrially
- In a silicon crystal, each atom shares electrons with 4 neighbours via covalent bonds
- The covalent lattice (face-centered cubic) leaves no free carriers in pure silicon at 0 K
- Next lecture: how heat, doping, and the PN junction turn this crystal into a device

Coming Up — Lecture 2

- **Title**
  - Conduction, Doping & the PN Junction
- **You Will Learn**
  - How discrete atomic energy levels merge into bands in a solid
  - How thermal energy generates electron-hole pairs and produces current
  - How doping creates N-type and P-type material
  - What happens at the boundary where P-type meets N-type — the PN junction
- **Required Reading**
  - Lecture Notes pp. 16–27

Thank You — Questions?